《Smart Mobile Application Development》

Software Design Document

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1. Introduction

Waynote is a comprehensive Jetpack Compose-based Android application designed to streamline the entire travel experience from discovery to booking. It integrates several key features, including travel discovery, trip planning, mock flight booking, community engagement, real-time chat, and the ability to save favorite destinations and posts, all in one application.

1.1 Target Audience

Waynote is tailored to modern Android users (API 24+), optimized for devices running API 26+ with the use of java.time for advanced date handling. The app employs a sleek, light blue Material 3 theme to ensure a visually appealing user interface. The app is designed with offline capabilities to retain important data locally, ensuring uninterrupted user experience even in the absence of a network connection.

1.2 Core Features

- **Travel Discovery:** Offers a rich catalog of destinations, with comprehensive information, images, and user reviews, all presented in a localized format (available in English and Chinese).
- **Trip Planning:** Users can plan their trips with template-driven or custom itineraries, save drafts, and revisit them for later edits.
- **Flight Booking:** The app demonstrates a mock flight booking system, allowing users to search for flights, sort results, and complete a mock checkout process.
- **Community Engagement:** A dedicated space for users to browse community posts, interact with others, share experiences, and follow topics of interest.
- **Real-Time Chat:** A messaging system that enables users to chat with contacts and store their conversation history.
- **Favorites:** Users can save destinations, posts, and other content to their personal collection for easy access later.

2. Requirements Specification

2.1 User Needs

- Browse through various destinations with English and Chinese.
- Create, modify, and save trip plans with templates and drafts.
- Engage in mock flight booking, maintaining user-selected flights in an order list.
- Browse community posts, follow favorite content, and interact via real-time chat.
- Manage personal profile, including avatar customization, language settings, and saved content.
- Simple authentication system with the option to skip or lock the user out after 3 unsuccessful attempts.

2.2 Functional Requirements

- **Authentication:** Login and registration flow with a 3-attempt lockout and optional skip.
- **Navigation:** Bottom navigation that includes Home, Messages, and Profile with state restoration.
- **Destination Catalog:** Includes favorite toggles for destinations and posts, a community feed (recommended, following, activities), and search functionality.
- **Flight Booking:** Mock flight search, result sorting, expanded information, passenger selection, and checkout success.
- **Trip Planner:** Trip creation with date validation, auto-fill templates, manual stops, and the ability to save trips.
- **Profile Management:** Users can customize their profile with avatar picking, language toggle, and view shortcuts to their orders and favorites.

2.3 Non-functional Requirements

- **Performance:** Optimized for fast load times and minimal data consumption.
- **Offline Functionality:** The app works offline to retain key data such as saved trips, orders, and favorites.
- **Accessibility:** Ensure accessibility for all users, including contrast ratios for readability, dynamic text support, and talkback labels for blind or visually impaired users.

3. Overall Design

3.1 Architecture

The app uses a single-activity architecture with NavHost for navigation. Routes are managed by a set of destination constants and a bottom-tab enum that provides easy access to key areas. Navigation is argumentized, allowing deep linking for specific destinations or chat sessions.

3.2 State Management

- **DataStore** is used for storing preferences like authentication status, avatar images, trip plans, favorites, orders, and chat history.
- **Compose's rememberSaveable** is utilized for UI state, ensuring a smooth experience during orientation changes.
- Complex objects like orders and chat data are encoded/decoded using JSON and CSV helpers to persist data offline.

3.3 Feature Modules

- **Authentication:** Handles login, registration, and state restoration on re-entry.
- **Home:** Displays featured destinations, promotional content, and popular picks.
- **Community:** Offers a tabbed community feed with media-rich posts and interactive elements such as bookmarking and posting.
- **Messages/Chat:** Supports real-time messaging with thread lists, unread badges, avatars, and persisted chat history.

- **Trip Planning:** Offers templates for quick trip creation and manual entry, along with a saved trip view.
- **Flight Booking:** Simulates a booking flow with a multi-step process for searching, selecting, and confirming flight choices.
- **Favorites:** Allows users to save their favorite destinations and posts.
- **Profile:** Enables users to manage their profile, including their avatar, language preference, and order/favorites management.

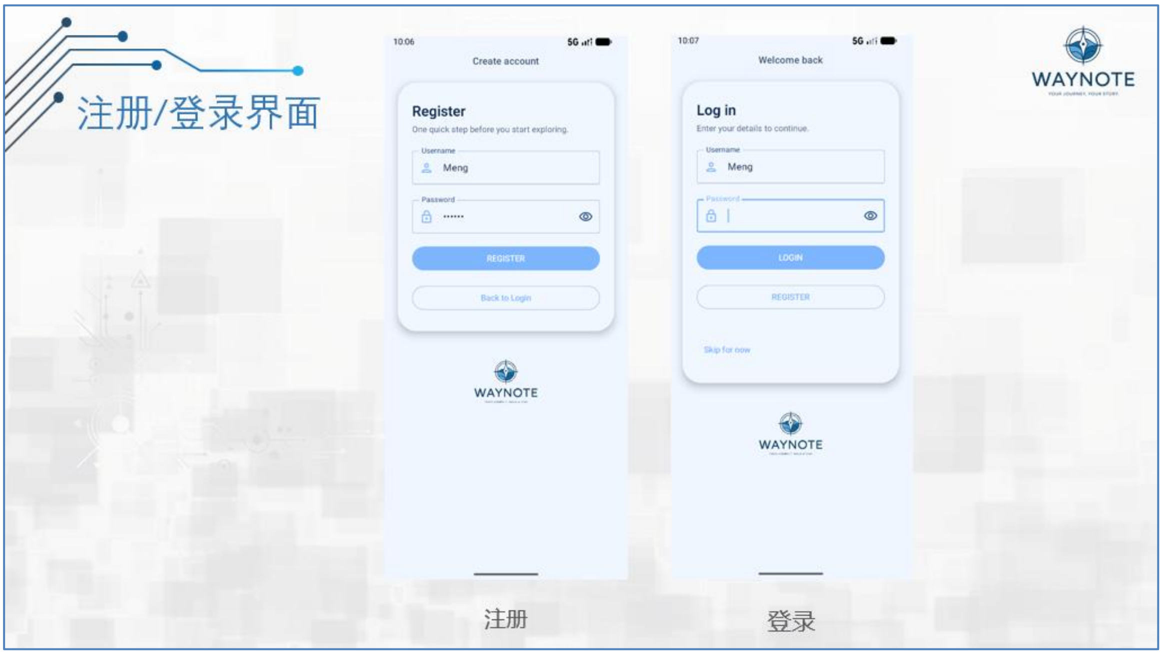
4. User Interface Design

4.1 Visual Language

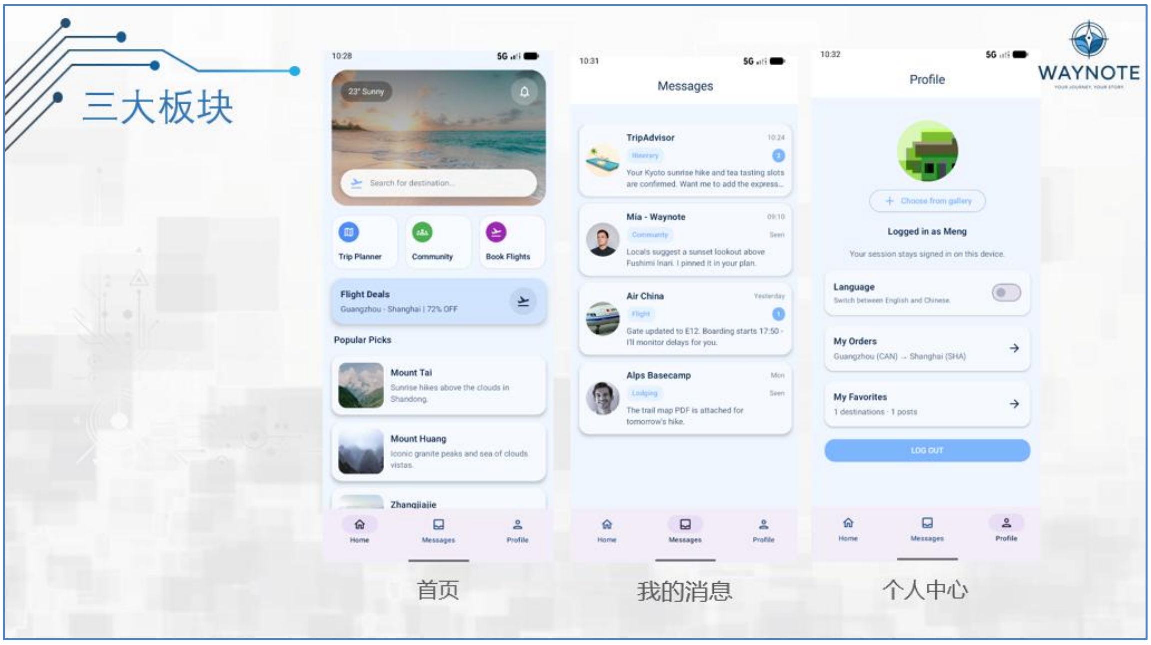
- **Color Palette:** The app uses a light blue palette with Aqua accent colors, ensuring a calm, modern, and visually appealing design.
- **Typography:** Consistent use of Material 3 typography for clarity and readability. Headlines, body text, and captions are styled using the standard Material design principles.
- **UI Components:** The interface is built using rounded cards, shadows, and gradients on hero sections to create a more dynamic and modern feel. Material 3 components like top app bars, bottom navigation, sheets, and forms are used to maintain consistency.

4.2 Key Screens

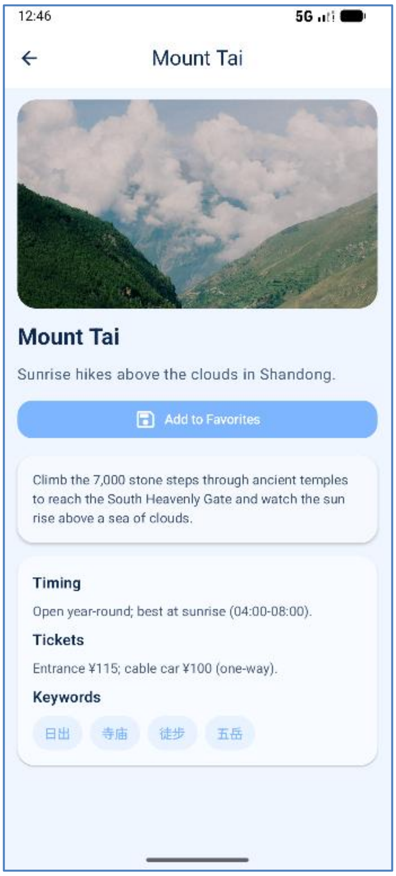
Login/Register and Skip Login:

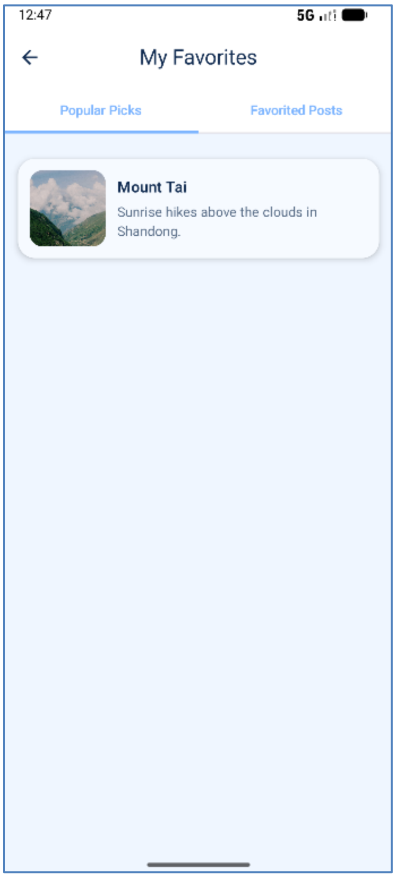


Homepage (banners, message reminders, section shortcuts, popular destinations):

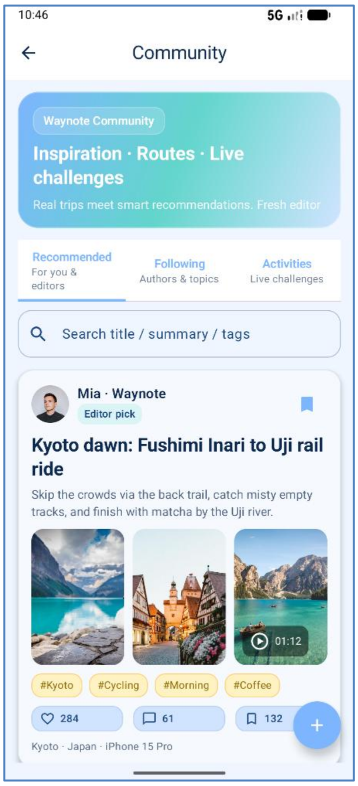


Destination details and bookmarks:

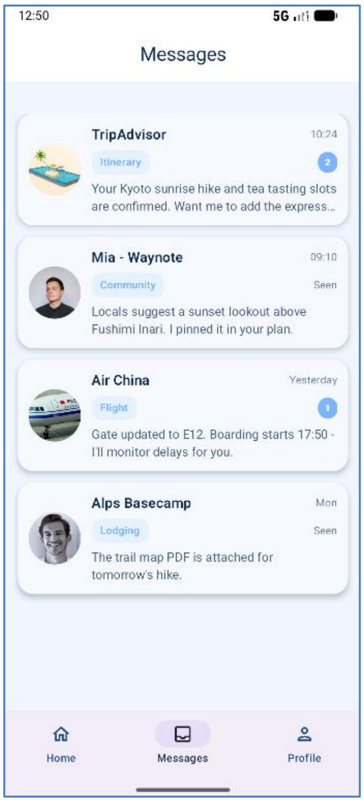


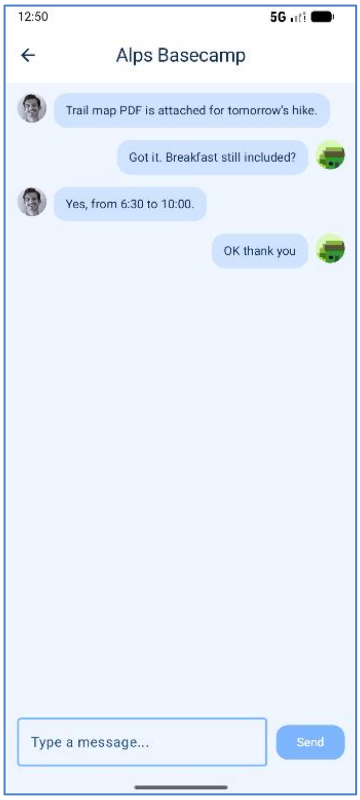


Community multi tab (recommended/followed/active), post/video display and collection:

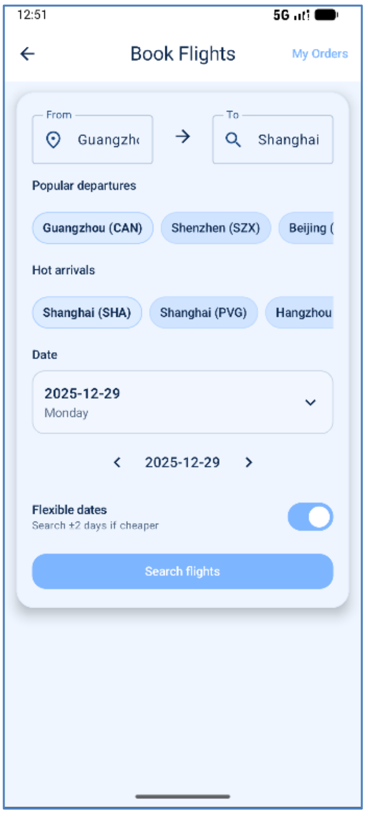


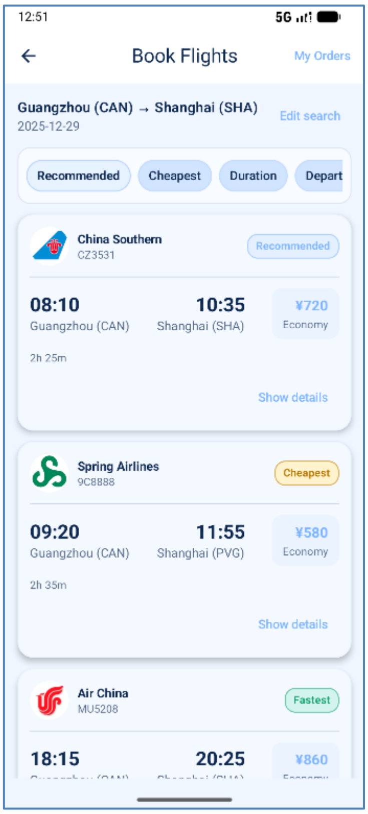
Message list and chat history:

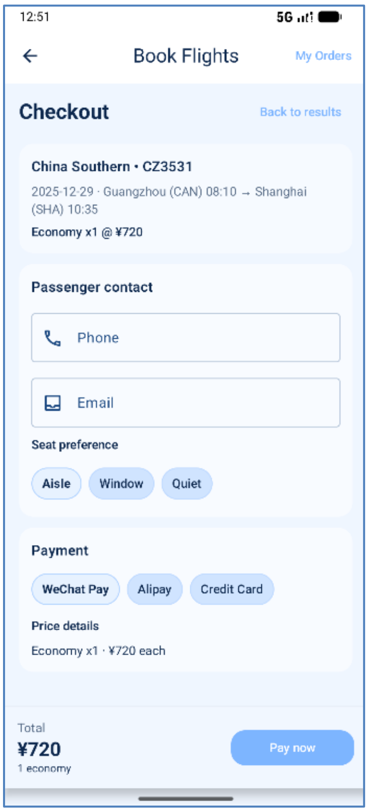


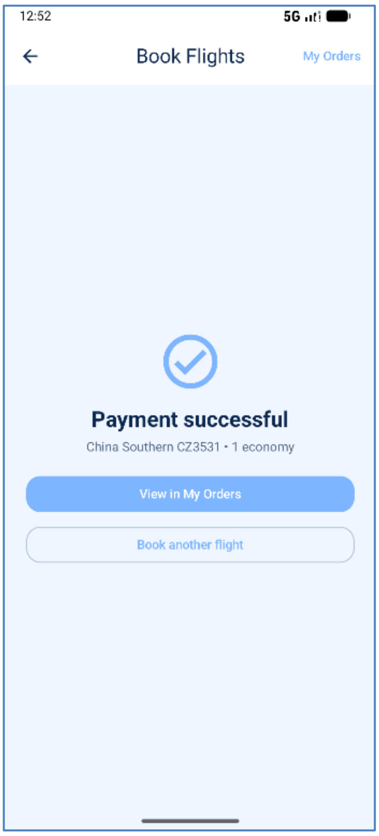


Flight ticket search - results - settlement - successful ticket issuance process and order list:



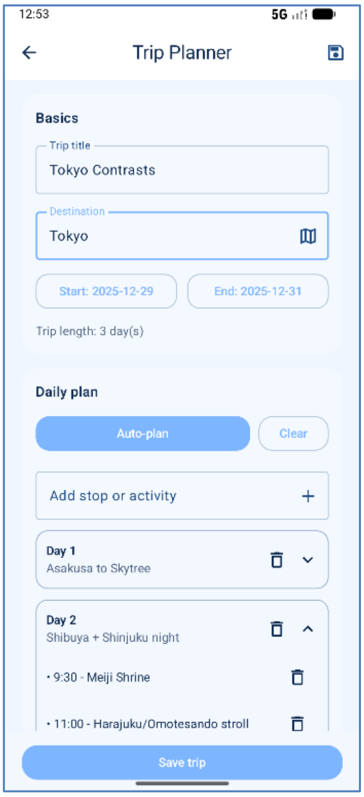




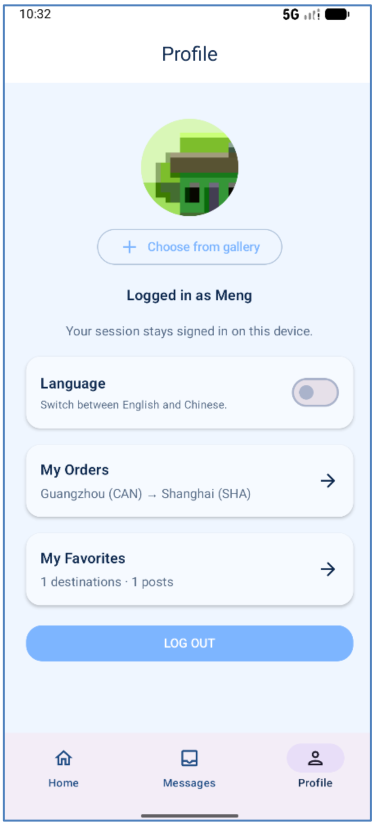


WAYNOTE

Schedule (automatic) planning editing and template saving:



Personal Center (avatar, language switching, entrance collection, logout):



4.3 Accessibility

- **Contrast and Colors:** High contrast between text and background to ensure readability, especially in bright or low-light environments.
- **Dynamic Text Support:** The app supports dynamic type sizes, making it accessible to users with visual impairments.
- **Touch Targets:** Larger touch targets ensure the app is easy to use on a variety of devices, especially for users with motor impairments.

5. Key Technologies & Challenges

5.1 Technologies

1. **Jetpack Compose + Material 3** for building modern and responsive UIs.
2. **Navigation-Compose** for routing, handling dynamic and static screens.
3. **DataStore Preferences** for efficient offline data storage.
4. **java.time** for date handling, ensuring robust support for date validations and calculations.
5. **Coil** for remote image loading and displaying visuals from URLs.

5.2 Challenges

1. Ensuring a seamless UX during lockout states and providing clear messaging to the user.
2. Date validation in trip planning, especially for users with older Android versions (pre-API 26).
3. Handling state restoration across app screens, ensuring user data persists when switching between tabs.
4. Realistic mock booking without backend support, maintaining flow realism without an actual flight API.

6. Testing and User Experience Analysis

6.1 Third-Party Cloud Testing Results

WeTest was used as the third-party cloud testing platform for this round. A total of 20 devices were covered in the test, including 1 device that was not tested. The devices came from various brands, such as Huawei, OPPO, vivo, and Samsung. All the tested devices passed successfully with a 100% success rate. The devices featured different hardware configurations and Android versions, ranging from Android 9 to Android 13.

6.1.1 Device Specifications

The tested devices included Huawei (e.g., Huawei Mate 30, Huawei P10, Huawei Y100), OPPO (e.g., OPPO Reno4 Pro, OPPO Reno2 Z), vivo (e.g., vivo Y100, vivo V1831A), and Samsung (e.g., Galaxy Note 20 Ultra, Galaxy S20+ 5G) models. The system versions included Android 9, 10, 11, and 12, with some devices using custom ROMs. The hardware configurations also varied, with different CPUs (e.g., Qualcomm Snapdragon, MediaTek Helio, Kirin 990) and memory configurations (ranging from 4GB to 12GB).

6.1.2 Compatibility Conclusion

1. **Overall Pass Rate:** All 19 devices tested successfully passed the tests, with a success rate of 100%. No issues were found in the report.
2. **Problematic/Failed Device List:** Any issues such as crashes, black screens, freezes, ANRs, permission/compatibility errors, etc., typically fall under the "compatibility failure device distribution/problem list."
3. **Untested Device:** This report includes one device marked as "untested" (Huawei Mate 30, system version 4.0.0). It is assumed that the issue was caused by device scheduling/scripts/environment.

兼容成功设备分布												
设备ID	设备品牌	硬件型号	机型名称	系统版本	内存(MB)	分辨率	CPU核数	CPU名称	CPU频率	CPU架构	结果	设备详情
2744	HUAWEI	ARS-AL00	华为畅享MAX	9	4096	1080x2244	8	高通 骁龙660	1.9	Kryo 260 Gold & 4x1	通过	353782/device/2744
8666	OPPO	PDNM00	OPPO Reno4 Pro	10	8192	1080x2400	8	高通 骁龙765G	1.8	& 1x2.2 GHz Kryo 4	通过	353782/device/8666
1001	HUAWEI	AQM-AL10	荣耀 Play 4T Pro	3.0.0	6144	1080x2400	8	麒麟 810 (7 nm)	1.9	Hz Cortex-A76 & 6x1	通过	353782/device/1001
8719	HUAWEI	POT-AL00a	华为 Enjoy 9s	2.0.0	6144	1080x2340	8	麒麟 710 (12 nm)	1.7	Hz Cortex-A73 & 4x1	通过	353782/device/8719
11295	realme	RMX3478	真我 Q5	13	6144	1080x2412	8	通 SM6375 骁龙 695	1.9	Kryo 660 Gold & 6x1	通过	53782/device/11295
11183	vivo	vivo 1907	vivo S1	11	6144	2340x1080	8	发科 MT6768 Helio F	1.7	Hz Cortex-A75 & 6x1	通过	53782/device/11183
12317	HUAWEI	MXW-AN00	荣耀 30 Youth	10	6144	1080x2400	8	MT6873V 天玑 800	2	Hz Cortex-A76 & 4x2	通过	53782/device/12317
13060	vivo	V2313A	vivo Y100	14	8192	2400x1080	8	发科 天玑 900 (6 nr	1.8	ARMv8a	通过	53782/device/13060
46	vivo	V1831A	vivo S1	10	6144	1080x2340	8	联发科 Helio P70T	2	Hz Cortex-A73 & 4x2	通过	00853782/device/46
7892	vivo	V2166A	vivo Y33s	12	8192	1600x720	8	斗 MT6833 天玑 700	2	Hz Cortex-A76 & 6x2	通过	353782/device/7892
11642	OPPO	PCKM80	OPPO Reno2 Z	10	8192	2340x1080	8	联发科 HelioP90	2	Hz Cortex-A75 & 6x2	通过	53782/device/11642
664	HUAWEI	OXF-AN10	荣耀 V30 Pro	10	8192	1080x2400	8	麒麟 990 5G (7 nm+)	2	6 & 2x2.36 GHz Corte	通过	0853782/device/664
2711	HUAWEI	VTR-AL00	华为 P10	9	4096	1080x1920	8	麒麟 960 (16 nm)	2.362	Hz Cortex-A73 & 4x1	通过	353782/device/2711
5382	HONOR	STK-LX1	荣耀 9X	9	4096	1080x2340	8	麒麟 710F (12 nm)	1.8	Hz Cortex-A73 & 4x1	通过	353782/device/5382
13064	realme	RMX2061	真我 6 Pro	11	8192	1080x2400	8	高通 骁龙 720G	1.9	Kryo 465 Gold & 6x1	通过	53782/device/13064
13533	OPPO	CPH1931	OPPO A5	10	3072	720x1600	8	高通 骁龙665	1.9	Kryo 260 Gold & 4x1	通过	53782/device/13533
124	vivo	V1950A	vivo NEX 3S	10	8192	1080x2256	8	高通 骁龙865	1.8	7 & 3x2.42 GHz Corte	通过	0853782/device/124
13603	SAMSUNG	SM-N9860	alaxy Note 20 Ultra 5	13	12288	3088x1440	8	高通 骁龙865 Plus	1.8	M5 & 2x2.50 GHz Co	通过	53782/device/13603
13165	SAMSUNG	SM-G9860	Galaxy S20+ 5G	10	12288	1440x3200	8	高通 骁龙865	1.9	M5 & 2x2.50 GHz Co	通过	53782/device/13165

兼容失败设备分布														
未测试														
设备ID	设备品牌	硬件型号	机型名称	结果	问题类别	问题描述	系统版本	内存(MB)	分辨率	CPU核数	CPU名称	CPU频率	CPU架构	设备详情
13491	HUAWEI	TAS-AL00	华为 Mate 30	未测试			4.0.0	8192	2340x1080	8	麒麟 990 (7 nm+)	1.8	6 & 2x2.09 GHz Cort	3782/device/13491

6.1.3 Coverage Scope

1. **Brand Distribution:** The tested devices were primarily from Huawei (6), vivo (5), OPPO (3), realme (2), SAMSUNG (2), and HONOR (1).
2. **System Version Distribution:** The test covered major Android versions (e.g., 9, 10, 11, 12, 13, 14), and versions like 2.0.0, 3.0.0, and 4.0.0 were also found, typically seen in specific ROMs/ecosystems.
3. **Hardware Range:** The devices tested ranged from 3GB to 12GB of memory, and screen resolutions from 720p to 1080p.

6.2 Performance Metrics

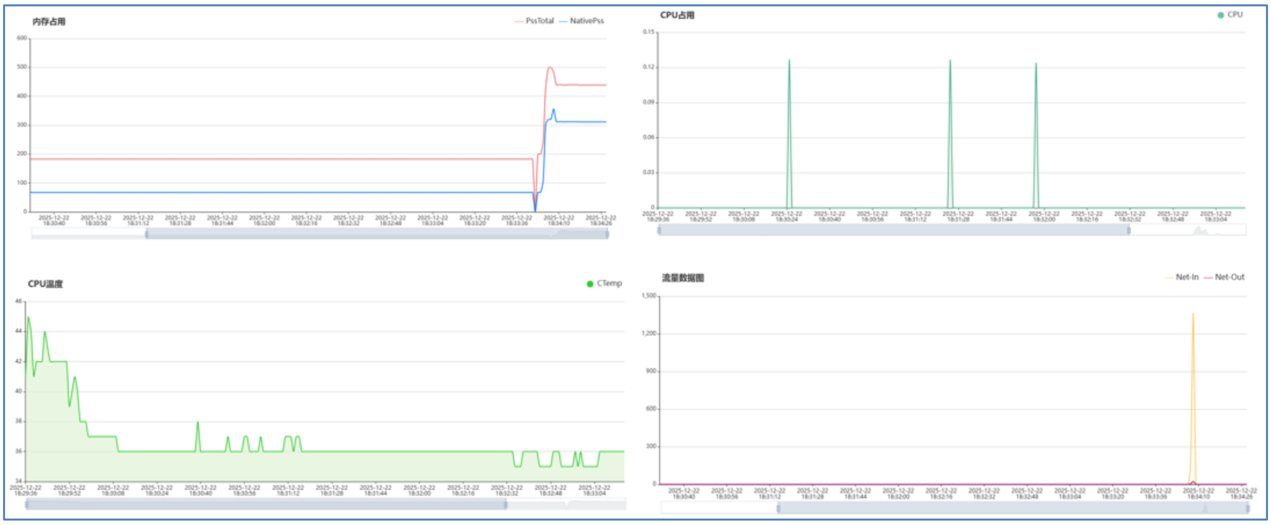
性能测试数据											
设备ID	设备品牌	机型名称	硬件型号	安装耗时(秒)	启动耗时(秒)	CPU占用(%)	cpuTemp2	内存占用(MB)	流量消耗(KB)	FPS(帧/秒)	设备详情
46	vivo	vivo S1	V1831A	8.48	1.51	0.07	/	183.16	0.00	0.28	0853782/device/46
124	vivo	vivo NEX 3S	V1950A	4.34	/	0.06	35.70	218.85	0.00	0.28	353782/device/124
664	HUAWEI	荣耀 V30 Pro	OXF-AN10	3.95	0.53	0.01	32.19	202.76	0.00	0.00	353782/device/664
1001	HUAWEI	荣耀 Play 4T Pro	AQM-AL10	7.56	0.72	0.11	35.29	181.00	0.00	0.72	53782/device/1001
2711	HUAWEI	华为 P10	VTR-AL00	7.02	1.16	0.13	28.95	107.68	0.00	0.23	53782/device/2711
2744	HUAWEI	华为畅享MAX	ARS-AL00	7.48	1.71	0.22	36.71	205.33	1523.84	0.33	53782/device/2744
5382	HONOR	荣耀 9X	STK-LX1	6.92	1.80	0.11	38.28	183.83	0.00	0.32	53782/device/5382
7892	vivo	vivo Y33s	V2166A	3.48	1.05	0.04	/	159.96	0.00	0.12	53782/device/7892
8666	OPPO	OPPO Reno4 Pro	PDNM00	8.62	0.56	0.02	32.09	206.40	0.00	0.00	53782/device/8666
8719	HUAWEI	华为 Enjoy 9s	POT-AL00a	12.73	1.50	0.09	44.31	180.22	0.00	0.24	53782/device/8719
11183	vivo	vivo S1	vivo 1907	6.75	1.30	0.07	/	123.20	0.00	0.16	3782/device/11183
11295	realme	真我 Q5	RMX3478	6.81	0.89	0.06	28.85	179.05	0.00	0.34	3782/device/11295
11642	OPPO	OPPO Reno2 Z	PCKM80	17.71	0.93	0.46	/	209.85	1521.53	0.44	3782/device/11642
12317	HUAWEI	荣耀 30 Youth	MXW-AN00	6.68	1.09	0.04	/	185.99	0.00	0.21	3782/device/12317
13060	vivo	vivo Y100	V2313A	4.46	1.10	0.05	34.31	174.35	0.00	0.28	3782/device/13060
13064	realme	真我 6 Pro	RMX2061	4.06	0.85	0.16	31.00	222.94	0.00	0.55	3782/device/13064
13165	SAMSUNG	Galaxy S20+ 5G	SM-G9860	4.22	0.30	0.19	29.07	226.09	1469.40	0.40	3782/device/13165
13491	HUAWEI	华为 Mate 30	TAS-AL00	2.31	0.56	0.05	36.07	141.54	0.00	0.21	3782/device/13491
13533	OPPO	OPPO A5	CPH1931	9.59	1.46	0.05	31.67	195.15	0.00	0.06	3782/device/13533
13603	SAMSUNG	laxy Note 20 Ultra 5	SM-N9860	5.45	0.34	0.00	30.15	186.37	0.00	0.00	3782/device/13603

Metric	Average	Range
Installation Time	6.93 seconds	2.31 - 17.71 seconds
Launch Time	1.02 seconds	0.30 - 1.80 seconds
CPU Usage	0.10%	0.00 - 0.46%
Memory Usage	183.69 MB	Peak: 226.09 MB
Network Consumption	1523.84 KB	Reasonable for cloud testing
Frame Rate (FPS)	0.28 FPS	0.00 - 0.72 FPS

6.2.1 Outliers / Abnormal Devices

1. **Slowest Installation:** OPPO Reno2 Z (approximately 17.71 seconds) stood out as an outlier.
2. **Slowest Launch:** HONOR 9X (approximately 1.80 seconds) had a slower launch time.
3. **Highest Memory Usage:** Samsung Galaxy S20+ 5G (approximately 226.09 MB) had the highest memory usage.

6.2.2 Complementary Data



6.3 User Feedback and Experience

6.3.1 Overall Feedback

The user interface (UI) of the app was generally considered intuitive, with most users navigating smoothly. However, some users provided suggestions for improvement. The most common suggestion was to optimize the application's response time, as some users experienced slight delays, particularly on lower-spec devices.

ID	反馈	建议
1	写帖子挺方便，但我不确定笔记保存到了哪里。	增加一个清晰的“全部笔记”入口，并在保存后给出“已保存”的确认提示/Toast。
2	界面看起来很干净，但有些按钮/图标对我来说不够直观。	为关键操作（新建、保存、同步、分享）添加文字标签或工具提示。
3	搜索很有用，但当有时拼错单词时，没有找到想要的内容。	支持模糊搜索，并高亮显示匹配到的关键词。
4	我刚开始用的第一分钟并不知道这个应用能做什么。	增加一个简短的首次运行引导（2–3 步）。
5	我想撤销一次修改，但不确定是否有撤销功能。	提供可见的撤销/重做控件，并展示键盘快捷键。
6	在手机上，一些元素显得拥挤，容易误触按钮。	增大可点击区域，并为小屏幕优化间距。
7	担心会丢失笔记，因为我不理解同步/离线的行为。	增加同步状态指示，并提供离线模式提示信息。
8	功能没问题，但打开较长笔记时感觉有点慢。	优化长笔记性能（例如：延迟渲染、加快加载）。
9	我想分享一条笔记，但没法很快找到分享/导出入口。	增加导出/分享按钮，提供 PDF/Markdown/链接 等选项。
10	应用挺好，能记住我的偏好设置就更好了。	持久化偏好设置（主题、上次打开的笔记、排序方式）。

6.3.2 Device-Specific Issues

Some devices experienced slow app responsiveness, especially those with lower specs or older Android versions. For instance, Huawei Mate 30 (Android 10) and OPPO Reno2 Z (Android 9) had longer launch times. A few users reported intermittent freezing or delays when switching between different sections of the app, especially on devices with lower processing power.

6.3.3 Suggestions for Improvement

- Users suggested further optimization of the app to improve performance on lower-spec devices.
- There was a request for clearer visual indicators, such as loading bars or progress indicators, to improve user perception of the app's responsiveness.
- Users also expressed the need for better cloud service integration, ensuring smoother data synchronization and improving offline functionality.

7. Conclusion

Waynote provides a cohesive, localized travel experience with a well-designed UI and persistent state management. By integrating features like travel discovery, trip planning, mock flight booking, community engagement, and real-time chat, Waynote successfully creates an end-to-end travel workflow that meets the needs of modern Android users. The app ensures a seamless user experience, even when offline, by leveraging efficient data storage and local persistence.

7.1 Achievements

- **End-to-End Workflow:** The app delivers a complete travel experience, from discovering destinations and planning trips to booking flights and saving favorite content.
- **Bilingual UI:** With localization in both English and Chinese, the app reaches a broader audience, catering to diverse language preferences.
- **State Management:** Persistent state handling is implemented through DataStore, ensuring users' data such as trip plans, favorites, and chat history are retained even when offline.

- **Seamless User Experience:** Through the use of Jetpack Compose and Material 3, Waynote offers a smooth, responsive interface with modern design elements and a consistent theming system.

7.2 Challenges Encountered During Development

- **Backend Simulation:** One of the main challenges was simulating a real-world flight booking flow without integrating a backend. This was overcome by creating a mock system, though future versions would benefit from real data integration for a fully functional booking experience.
- **Device Compatibility:** Ensuring compatibility across a range of devices and Android versions (API 9 to API 13) was crucial. The app passed testing with a 100% success rate on 19 devices, though performance on lower-spec devices was slower.
- **State Restoration:** Another challenge was ensuring smooth state restoration, especially when switching between different app screens. This was addressed using Compose's rememberSaveable, although further improvements in state handling could be made with the MVVM architecture.

7.3 Suggestions for Future Improvements

- **Real Data Integration:** Integrating actual APIs for flights, destinations, and user data will enhance the app's functionality, providing users with up-to-date information. This will also require implementing offline caching and robust error handling to ensure a seamless user experience.
- **MVVM Architecture:** Moving to an MVVM architecture with ViewModel and repository separation will make the app more maintainable and scalable. This will simplify state management, allow better testing, and separate concerns more effectively across different parts of the app.
- **Encrypted Storage and Enhanced Security:** Implementing encrypted storage for sensitive user data, such as authentication credentials and personal information, will improve security. Additionally, strengthening authentication flows, such as adding multi-factor authentication, will enhance user trust and data protection.
- **Improved Trip Planner:** Future versions should introduce a calendar export feature, enabling users to export their trips to their preferred calendar app. Additionally, incorporating notifications to remind users of upcoming trips and a richer template gallery for trip planning would enhance usability.
- **Accessibility Enhancements:** To ensure the app is accessible to a wider audience, a thorough accessibility review should be conducted. Improvements could include TalkBack labels for screen readers, larger touch targets for easier navigation, and support for dynamic text to cater to users with visual impairments.

In summary, Waynote has successfully delivered a localized, intuitive, and feature-rich travel app that enhances the way users plan and manage their trips. Moving forward, by integrating real-world data sources, optimizing performance for lower-spec devices, and implementing a more secure and scalable architecture, Waynote will continue to evolve into a powerful tool for travelers.